Custom Commands in `doug-test.sty`

A Reference for Workshop Participants

Douglas Tam

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This document lists every custom command defined in `doug-test.sty`, with a brief explanation of its purpose and its parameters. Diagram commands auto-scale to a consistent box (typically 5×5 units, or $\sim$4 cm for 3D shapes) so figures look uniform across a page.

Conventions. Optional arguments are shown in `[brackets]`; mandatory arguments in {`braces`}. Distances default to millimetres unless noted.

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1 Spacing and Answer Helpers

`\svert[1]`

Purpose: Vertical stretch filler. Pushes subsequent content down the page proportionally; use to reserve working space between questions.

Parameters: `[1]` = stretch factor (default `1`). Larger values reserve more space.

`\VSP[3]`

Purpose: Fixed vertical space in millimetres. Useful for small, predictable gaps (e.g. between subparts).

Parameters: `[3]` = space in mm (default `3`).

`\ansbox[1]`

Purpose: Right-aligned "Ans:" line with a rule for the final answer.

Parameters: `[1]` = stretch factor for vertical space above the line (default `1`).

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2 2D Diagram Commands

`\drawangle[0.8]{A}{B}{C}`

Purpose: Labels the angle at vertex B in triangle ABC, computing the measure automatically via `tkz-euclide`.

Parameters: `[0.8]` = label distance from vertex; `{A}{B}{C}` = three points (middle argument is the vertex).

`\drawright[0]{h}{v}`

Purpose: Right triangle with horizontal leg h and vertical leg v, auto-scaled to fit a 5×5 box.

Parameters: `[0]` = rotation angle in degrees; `{h}` = horizontal leg; `{v}` = vertical leg.

`\labelside[opts]{P}{Q}{text}`

Purpose: Places a label at the midpoint of segment PQ (typically a side length).

Parameters: `[opts]` = TikZ node options (e.g. `above`, `below`); `{P}{Q}` = endpoints; `{text}` = label.

`\drawanytriangle[0]{a}{b}{c}`

Purpose: Triangle with side lengths a, b, c; validates the triangle inequality; auto-scales to a 5×5 box. Prints "Invalid triangle!" if the sides are inconsistent.

Parameters: `[0]` = rotation angle; `{a}{b}{c}` = three side lengths.

`\drawquad[blue]{a}{h}{k}{xmin}{xmax}`

Purpose: Plots $y = a(x - h)^2 + k$ over a domain, smooth with arrowheads.

Parameters: `[blue]` = colour; `{a}` = leading coefficient; `{h}` = horizontal shift; `{k}` = vertical shift; `{xmin}{xmax}` = domain.

`\drawaxis{xmin}{xmax}{ymin}{ymax}`

Purpose: Draws x- and y-axes with arrowheads and labels, clipped to the window with a small buffer.

Parameters: `{xmin}{xmax}{ymin}{ymax}` = coordinate window.

`\drawgaxis{xmin}{xmax}{ymin}{ymax}`

Purpose: Same as `\drawaxis`, plus a faint background grid.

Parameters: Same four parameters as `\drawaxis`.

`\setupdivision{divisor}{dividend}`

Purpose: Formats a long-division bracket: divisor outside, dividend inside an overline.

Parameters: {`divisor`} = expression outside the bracket; {`dividend`} = expression inside.

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3 3D Shape Commands

All 3D commands auto-scale to fit roughly a 4 cm box and accept a unit string as the final argument (e.g. `cm`, `m`, `in`).

`\drawrectprism{length}{width}{height}{unit}{scale}`

Purpose: Rectangular prism with labelled length, width, and height. Hidden edges dashed.

Parameters: {`length`}, {`width`}, {`height`} = dimensions; {`unit`} = unit label; {`scale`} = outer TikZ scale factor (use `1` for default).

`\drawtriprism{base}{triheight}{prismlength}{unit}`

Purpose: Isosceles triangular prism. Base, triangle height, and prism length labelled.

Parameters: {`base`} = base of triangular face; {`triheight`} = height of triangular face; {`prismlength`} = length of prism; {`unit`} = unit label.

`\drawrighttriprism{baseleg}{heightleg}{prismlength}{unit}`

Purpose: Right-triangle prism. Base leg, height leg, and prism length labelled.

Parameters: {`baseleg`}, {`heightleg`} = legs of the right triangular face; {`prismlength`} = length of prism; {`unit`} = unit label.

`\drawcylinder{radius}{height}{unit}`

Purpose: Vertical cylinder with radius and height labelled.

Parameters: {`radius`} = cylinder radius; {`height`} = cylinder height; {`unit`} = unit label.

`\drawcone{radius}{height}{unit}`

Purpose: Cone with radius and height labelled, right-angle marker at the base.

Parameters: {`radius`} = base radius; {`height`} = cone height; {`unit`} = unit label.

`\drawsphere{radius}{unit}`

Purpose: Sphere with diameter labelled (diameter computed automatically as $2r$).

Parameters: {`radius`} = sphere radius; {`unit`} = unit label.

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4 Front Pages

`\testfrontpage[title]`

Purpose: Full test cover: title box, rubric table for Understanding & Solving and Mathematical Communication with columns **0**–**6**, the school academic-integrity statement, and name/teacher lines.

Parameters: `[title]` = test title.

`\proftestpage[title]`

Purpose: Proficiency-style cover using the 5-band rubric (IE, Emerging, Developing, Proficient, Extending) for both outcomes. Ends with a page break.

Parameters: `[title]` = test title.

`\quizfrontpage[title]`

Purpose: Compact quiz cover: title box plus name and teacher lines. No page break.

Parameters: `[title]` = quiz title.

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5 Notes on Design

- **Auto-scaling.** Diagram commands normalise to a consistent box size, so figures across a test look uniform regardless of the raw side lengths or coordinates.
- **Validation.** `\drawanytriangle` checks the triangle inequality and warns the author ("Invalid triangle!") rather than drawing a degenerate figure.
- **Rubric transparency.** Rubrics are printed on the cover page by design, so students see the assessment criteria before they begin.
- **Multiple rubric models.** The package supports the 0–6 level model (`\testfrontpage`), the 5-band proficiency model (`\proftestpage`), and the quiz/notebook variants.